

KIRIROM INSTITUTE OF TECHNOLOGY*Department of Software Engineering***BLOCKCHAIN-BASED DIGITAL CERTIFICATE ISSUING PLATFORM**

Final Project Submission Report — Citadel Platform Architecture & Implementation

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Institution	Kirirom Institute of Technology (KIT)
Team Arrangement	Individual Project (100% Technical Contribution)
GitHub Repository	https://github.com/mengchheanglong/citadel-blockchain-certificates
Public Demo Video	Available on Google Drive / YouTube (Public Link)
Submission Date	October 2026

1. System Overview

1.1 The Credentialing Dilemma

Educational institutions and professional certifying organizations face growing challenges with traditional credential authenticity:

- **Vulnerability to Forgery:** Static PDF certificates and paper diplomas can easily be altered using graphic editing software or AI tools to modify recipient names, degree titles, or graduation years without visual detection.
- **Operational Bottlenecks:** Employers and university registrars must conduct manual verification via phone calls or email, typically requiring two to four weeks per verification.
- **Centralized Database Risks:** Centralized academic databases represent single points of failure susceptible to administrative corruption, insider tampering, and catastrophic data loss.

1.2 The Citadel Platform Solution

Citadel is a decentralized web application engineered to modernize credential issuing using Ethereum blockchain technology:

- **Cryptographic Immutability:** When a certificate is issued, a unique SHA-256 cryptographic fingerprint is permanently recorded on an Ethereum smart contract. Any alteration to student data alters the hash, immediately exposing the forgery.
- **Instant Zero-Gas Verification:** Verifiers authenticate credentials in seconds without requiring cryptocurrency, a Web3 wallet, or a platform account via free on-chain view queries.
- **Automated Issuance & Delivery:** The platform generates high-resolution vector PDF diplomas with embedded verification QR codes and dispatches them automatically to student inboxes via SMTP.
- **Lifecycle Governance:** Issuing bodies can assign validity periods (Lifetime, 1 Year, 2 Years) and revoke compromised certificates on-chain with mandatory audit justification logging.

2. System Architecture

Citadel utilizes a clean four-tier architectural separation of concerns designed for enterprise reliability, high throughput, and data privacy:

- **1. Presentation Layer (Frontend):** Next.js 14 App Router, TypeScript, and Tailwind CSS. Provides an administrative operations portal and a clean public verification engine.
- **2. Application & API Layer (Backend):** Next.js Server Route Handlers and Server Actions. Enforces Zod schema validation, computes deterministic SHA-256 digests, renders vector diplomas with jsPDF, and coordinates SMTP delivery via Nodemailer.
- **3. Persistence Layer (Database):** PostgreSQL managed via Prisma ORM 5.18. Stores recipient metadata, degree titles, and institution profiles off-chain to ensure GDPR/FERPA compliance.
- **4. Decentralized Ledger Layer (Blockchain):** Solidity 0.8.24 smart contract (CertificateRegistry.sol) deployed on the Ethereum EVM (Sepolia Testnet / Hardhat). Serves as the immutable registry for 32-byte credential hashes.

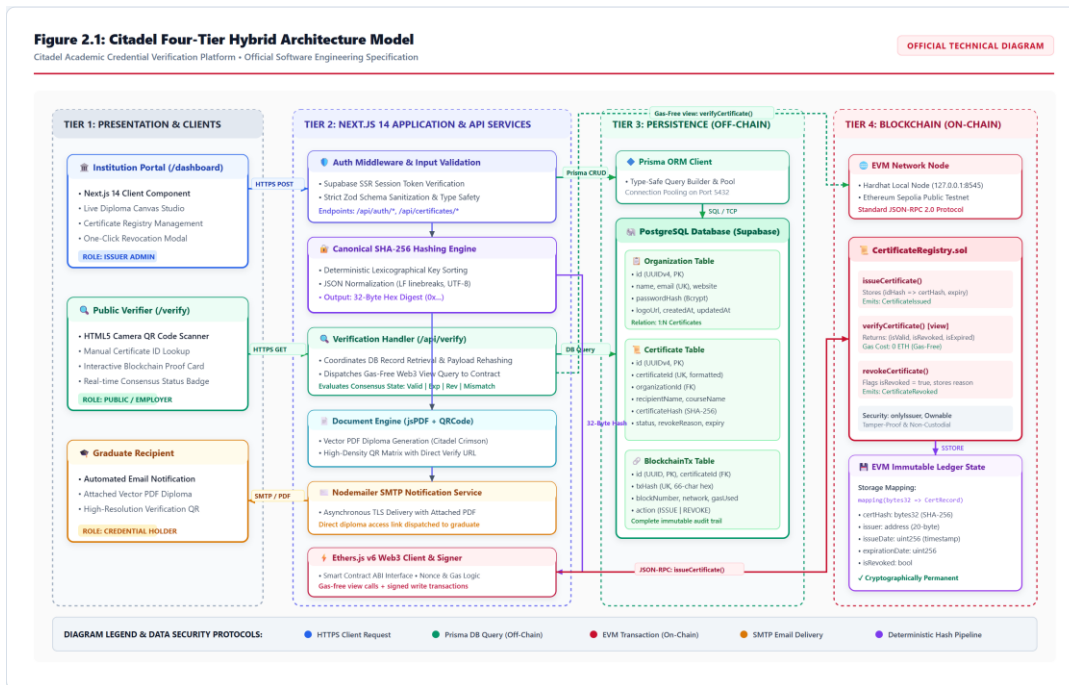


Figure 1: Citadel four-tier system architecture model.

3. User Flow / System Flow

The platform coordinates two primary workflows: the Organization Issuance Pipeline and the Public Verification Pipeline:

- **Organization Issuance Flow:** The accredited institution logs into the dashboard and accesses the Issue Studio. As the administrator inputs graduate information, an interactive vector diploma canvas updates in real time. Upon submission, the server computes a canonical SHA-256 fingerprint, calls `issueCertificate` on the smart contract, stores metadata in PostgreSQL, and generates an official PDF diploma emailed to the graduate.
- **Public Verification Flow:** An employer or admissions officer navigates to `/verify` and provides the credential via camera QR scan, image drop, PDF upload, or manual ID entry. The system retrieves the

record, re-computes the hash, executes a zero-gas view call on the smart contract, and renders an official cryptographic proof badge (Valid, Expired, or Revoked).

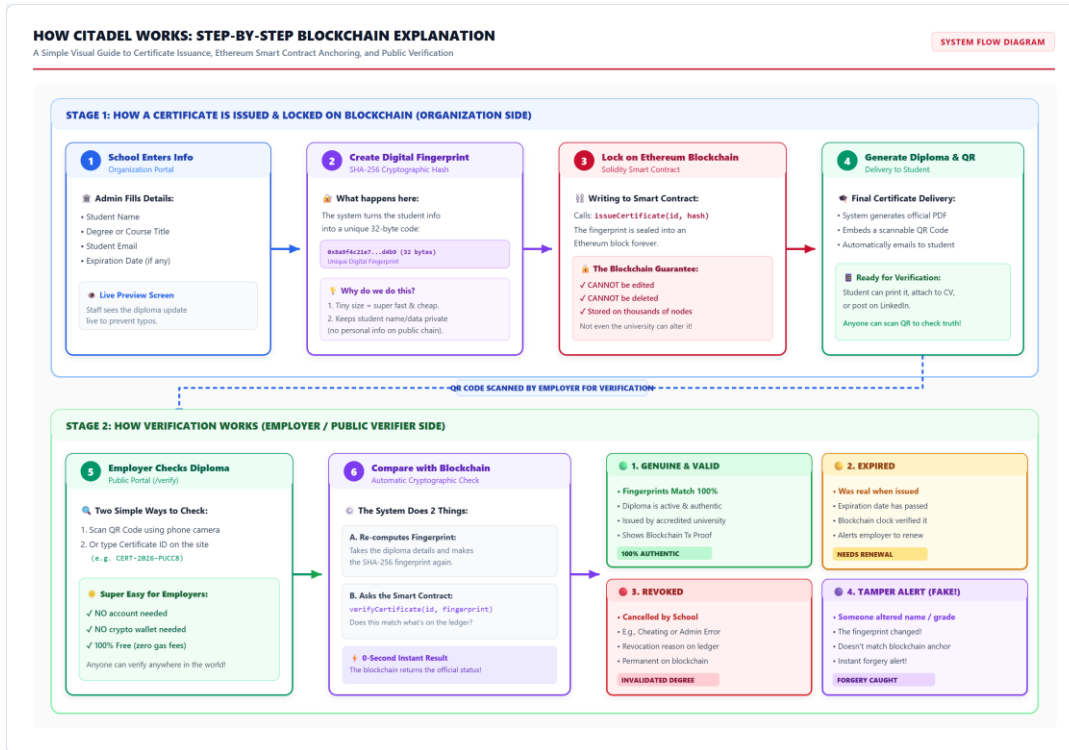


Figure 2: End-to-end credential lifecycle: Stage 1 Issuance Pipeline and Stage 2 Public Verification Pipeline.

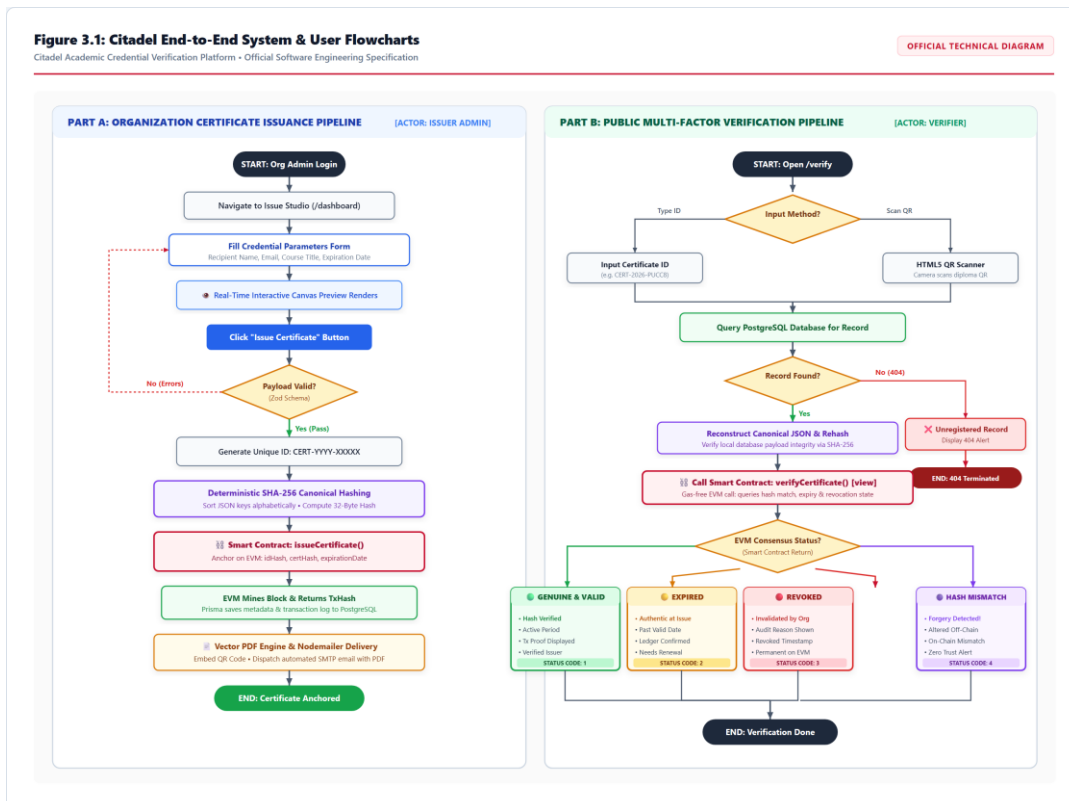


Figure 3: Technical sequence diagram illustrating interactions across User, Web Application, EVM Smart Contract, Database, and Recipient.

4. Database Design (ER Diagram)

To protect student privacy and minimize gas expenditure, detailed textual data is stored off-chain in PostgreSQL, with relational foreign keys connecting to on-chain transaction hashes:

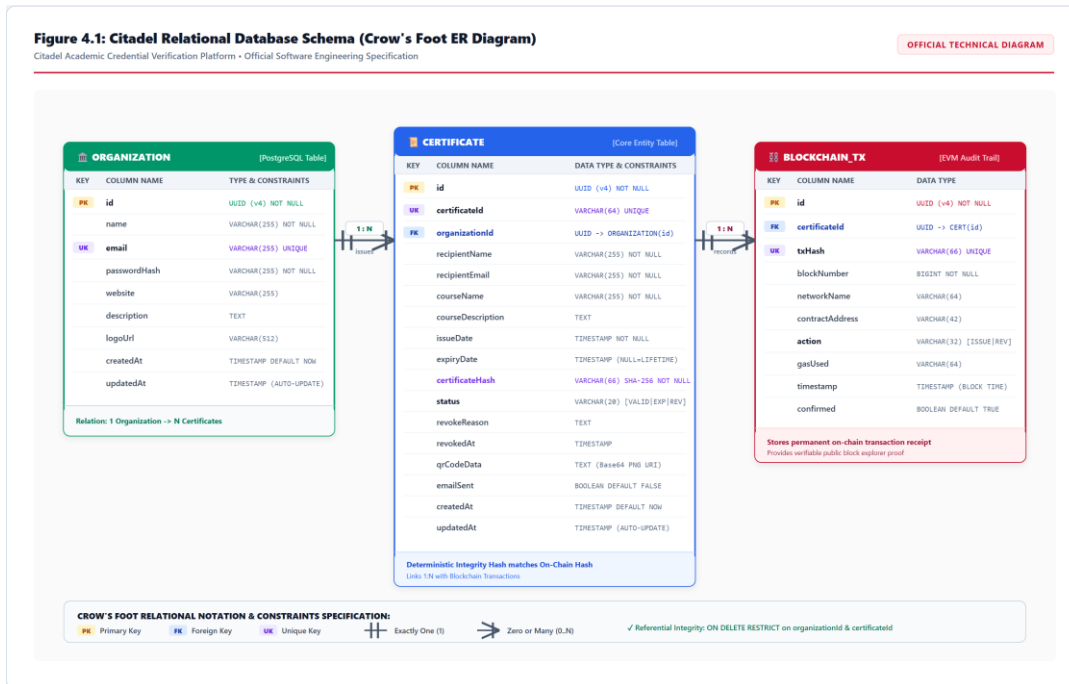


Figure 4: Entity-Relationship model representing Organization, Certificate, and BlockchainTransaction entities.

Table	Key Attributes & Constraints	Architectural Purpose
Organization	id (PK UUID), name, email (UK), website, description, createdAt	Stores verified institution profile and issuer credentials.
Certificate	id (PK UUID), certificateId (UK), recipientName, recipientEmail, courseName, issueDate, expiryDate, status, certHash, organizationId (FK)	Maintains academic records, expiration timestamps, and canonical SHA-256 digests.
BlockchainTransaction	id (PK UUID), certificateId (FK), txHash (UK), blockNumber, networkName, contractAddress, action, timestamp, confirmed	Audit log of on-chain Ethereum transaction receipts and block confirmations.

5. Blockchain Architecture

5.1 Hybrid On-Chain / Off-Chain Philosophy

Storing complete academic transcripts and personal identities directly on Ethereum is cost-prohibitive and violates privacy laws (GDPR Right to be Forgotten, FERPA student privacy). Citadel stores metadata in PostgreSQL while anchoring only a 32-byte cryptographic hash on-chain.

5.2 Deterministic Canonical Hashing Protocol

To prevent JSON key-ordering discrepancies across different programming platforms, Citadel sorts object keys lexicographically before calculating the SHA-256 digest:

Mathematical Cryptographic Specification: Canonical Payload = JSON.stringify(sortKeys({ certId, recipientName, courseName, issueDate, organizationId })))
 certHash = SHA-256(Canonical Payload)
 certIdHash = keccak256(certId)

Due to the avalanche effect of SHA-256, changing even a single byte in the graduate's name or course title produces a completely different hash, immediately causing on-chain verification to fail.

6. Smart Contract Design

The CertificateRegistry.sol smart contract is written in Solidity 0.8.24 and compiled with Hardhat:

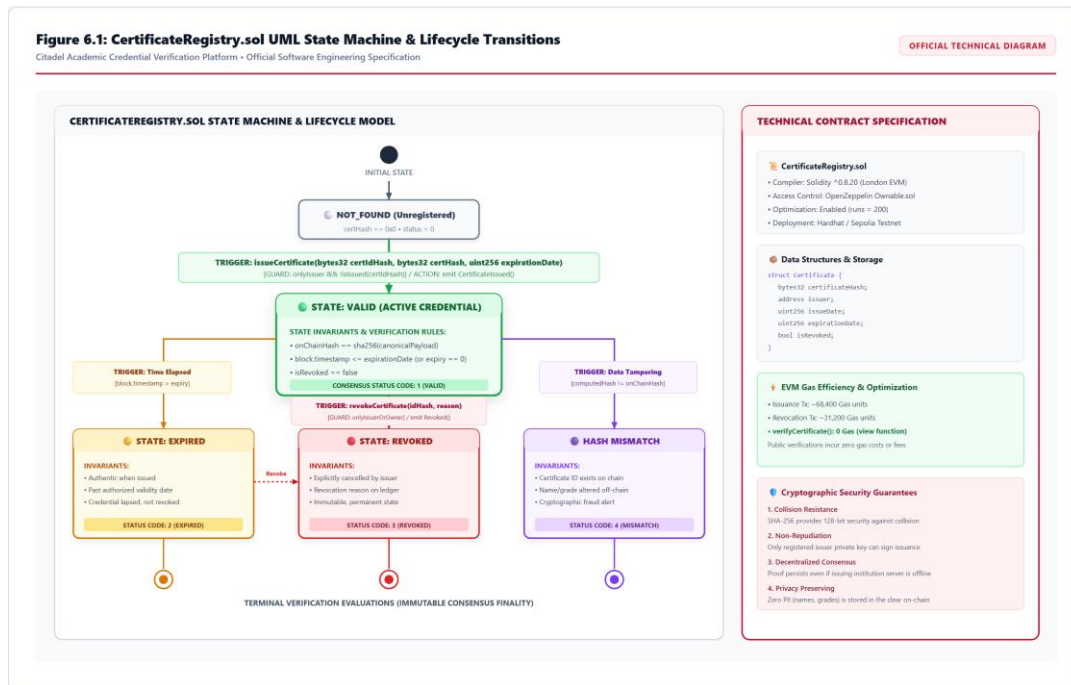


Figure 5: CertificateRegistry.sol Smart Contract Architecture: State Machine Lifecycle, Modifiers, and Cryptographic Invariance.

6.1 State Machine Lifecycle

- **Valid (1):** Credential exists on-chain, hash matches, and block.timestamp is before expirationDate.
- **Expired (2):** Credential is authentic, but the validity window has elapsed.
- **Revoked (3):** Formally invalidated on-chain by the issuing institution with recorded audit reason.
- **HashMismatch (4):** Certificate ID exists, but the presented data payload produces a mismatched hash.
- **NotFound (0):** Certificate ID has never been registered on-chain.

6.2 Key Smart Contract Functions

- **issueCertificate(bytes32 certIdHash, bytes32 certHash, uint256 expirationDate):** Enforces authorized issuer role, checks that the ID has not been used, verifies future expiry, and commits the certificate to the blockchain. Emits CertificateIssued.
- **verifyCertificate(bytes32 certIdHash, bytes32 certHash):** Free view function (zero gas). Compares the presented hash against the on-chain ledger and dynamically computes expiration.
- **revokeCertificate(bytes32 certIdHash, string reason):** Only the original issuing institution or contract administrator can revoke an active certificate. Emits CertificateRevoked.

7. User Interface Design or Screenshots

Below are the high-resolution screenshots demonstrating all required features across the Organization Portal and Certificate Verification Engine:

7.1 Organization Portal Interface

The Organization Portal provides accredited institutions with complete administrative control:

- **1. Organization Login:** Secure authentication with input validation and session routing.
- **2. Certificate Issuance:** Interactive split-screen studio where entered data renders a vector diploma canvas in real time.
- **3. Downloadable PDF Diplomas:** Automatic PDF creation with embedded QR code and SMTP email dispatch.
- **4. Blockchain Commitment:** Mined transaction confirmations with Tx Hash and Block Number on Ethereum Sepolia.
- **5. Issued Certificate Management:** Comprehensive registry table with status filters (All, Valid, Expired, Revoked) and CSV audit export.

The screenshot displays the 'Issue a certificate' interface within the Citadel Issuer Portal. The interface is divided into three main areas: a sidebar, a central form, and a live preview.

- Sidebar:** Contains navigation links for Overview (Dashboard), Credentials (Issue certificate, Registry), Organisation (Settings), Public Pages (Verification portal, Landing page), and Ledger (Local Hardhat node).
- Central Form:**
 - Registry > Issue a certificate:** The main heading.
 - Issue a certificate:** A sub-heading with a note: 'The details below are hashed together and written to the blockchain. They cannot be edited afterwards — only revoked and reissued.'
 - Recipient:** Fields for 'Full name' (Alexander J. Vance) and 'Email address' (alexander.vance@mit.edu). A note states: 'The PDF and verification link are sent here as soon as the transaction confirms.'
 - Award:** Fields for 'Programme or qualification' (Master of Science in Distributed Blockchain Systems) and 'Description or honours' (Awarded with Highest Distinction for thesis excellence in Cryptographic Zero-Knowledge Consensus Protocols).
- Live Preview:**
 - Kirirom Institute of Technology BLOCKCHAIN-VERIFIED CREDENTIAL:** The header of the diploma.
 - THIS IS TO CERTIFY THAT Alexander J. Vance HAS SATISFIED THE REQUIREMENTS FOR Master of Science in Distributed Blockchain Systems:** The main body of the diploma.
 - Issued:** 3 October 2026. **Valid until:** No expiry.
 - QR:** A QR code for verification.
 - Notes:** 'The delivered PDF follows this layout at A4, with the verification QR code generated once the certificate is anchored on-chain.'

Figure 6: Split-screen Issue Studio: form fields on left and real-time vector diploma preview on right.

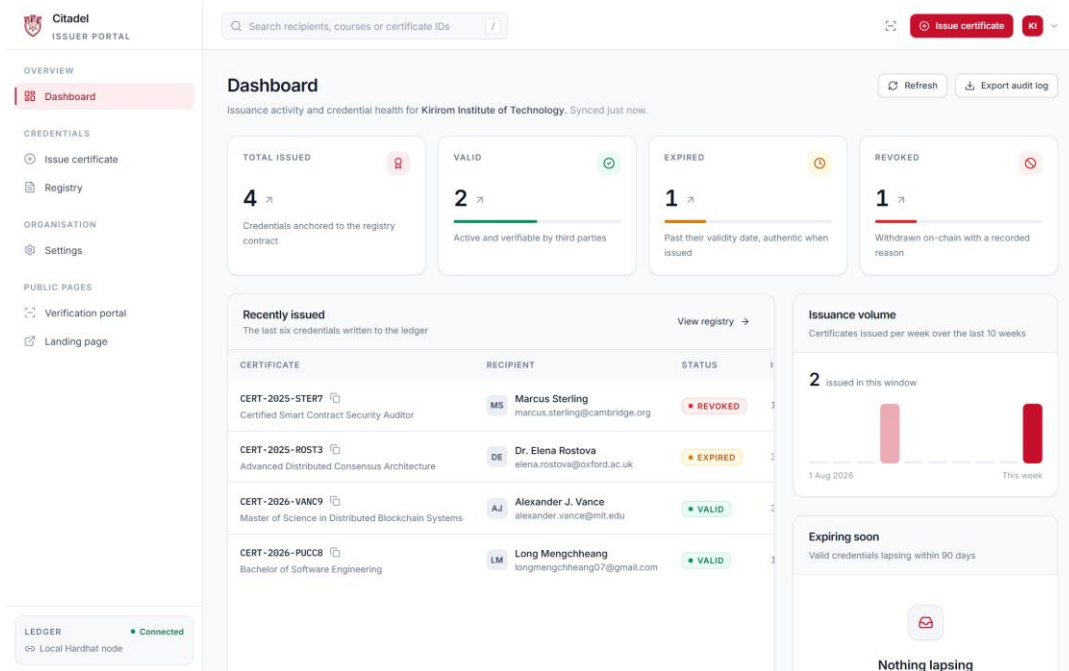


Figure 7: Organization Executive Dashboard showing live metrics, issuance trends, and recent records.

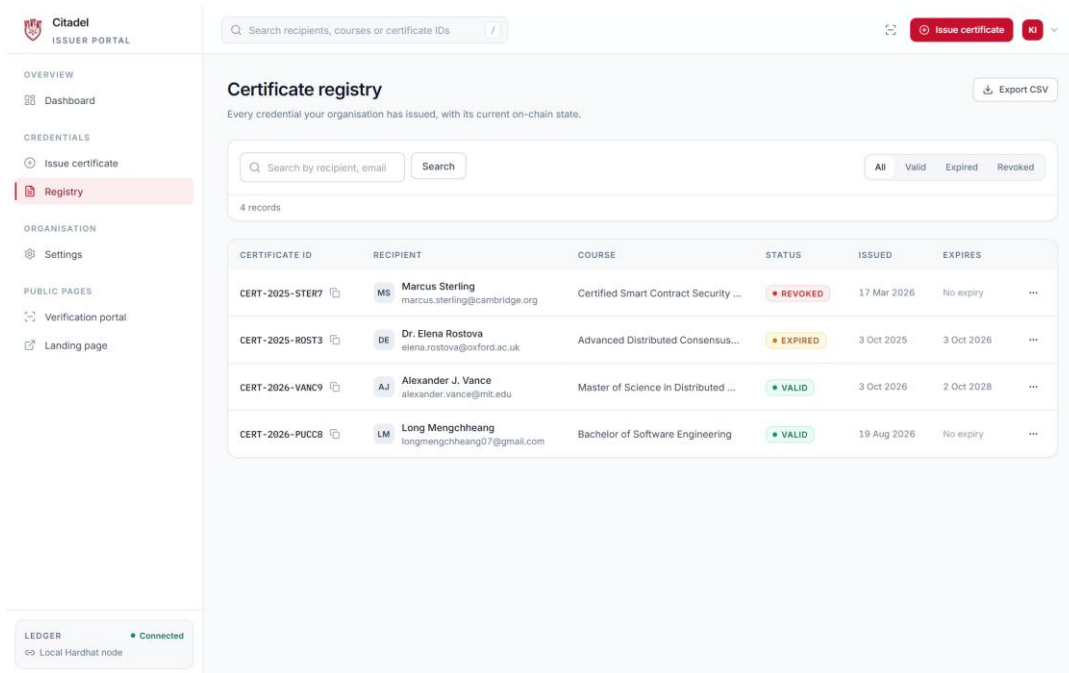


Figure 8: Certificate Registry table with status filter tabs and client-side CSV spreadsheet export.

7.2 Certificate Verification Interface

Public users verify credentials without accounts, gas, or Web3 wallets:

- 1. **Multi-Modal Verification:** Public users can verify via Certificate ID, camera QR scan, image drop, or PDF upload.
- 2. **Certificate Metadata Display:** Displays student name, degree title, issue date, and issuing accredited institution.

- **3. Blockchain Transaction Proof:** Displays on-chain transaction hash, block number, contract address, and Etherscan link.
- **4. Three Lifecycle Statuses:** Evaluates state across all three required statuses: Valid (Green), Expired (Yellow), and Revoked (Red).

Figure 9: Public verification portal supporting manual ID search, camera scanning, and file upload.

Figure 10: Cryptographic proof result: Valid status with on-chain Ethereum transaction confirmation.

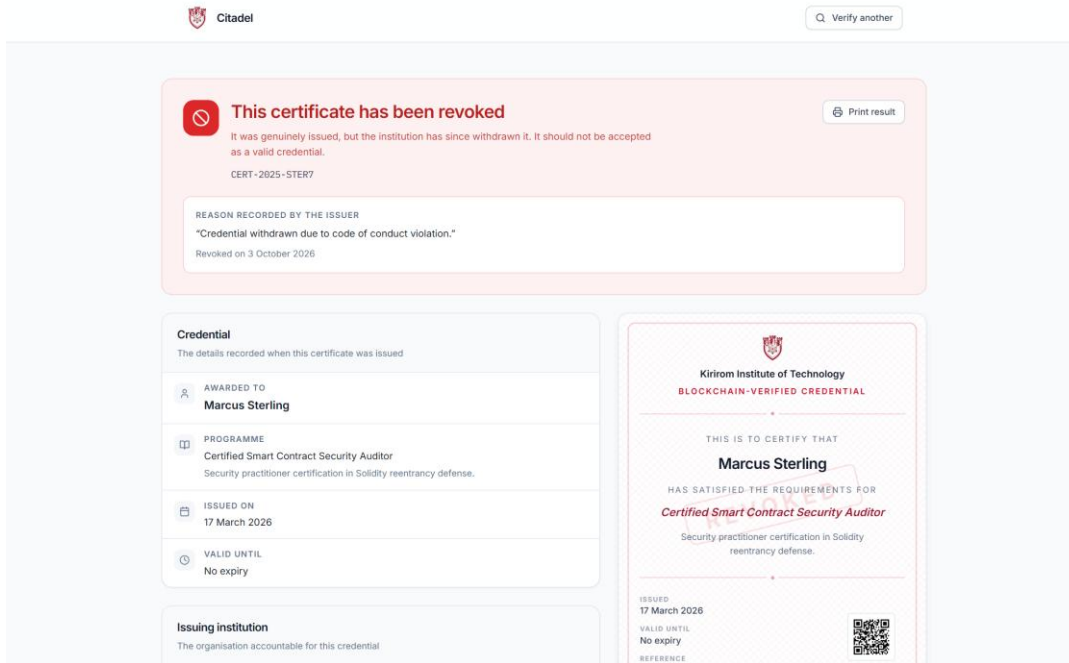


Figure 11: Revoked verification result: red status alert with recorded issuer audit reason and timestamp.

8. Implementation Summary

Citadel is implemented using modern full-stack web and blockchain technologies:

- **Frontend Stack:** Next.js 14.2 App Router, TypeScript, Tailwind CSS, Radix UI Primitives, Lucide Icons.
- **Backend Stack:** Node.js 20+, Next.js Server Actions & API Routes, Zod schema validation.
- **Database:** PostgreSQL (Supabase), Prisma ORM 5.18.0.
- **Blockchain Ledger:** Solidity 0.8.24, Hardhat 2.22.6, Ethers.js v6.13.1 (Sepolia Testnet / Hardhat EVM).
- **Document Services:** jsPDF 2.5.1 (vector diplomas), Nodemailer 6.9.14 (SMTP notifications).
- **Multi-Modal Scanner:** Html5Qrcode camera controller, jsQR multi-pass canvas engine, Node zlib decompression.

8.1 Automated Test Suite Results

The implementation was validated using a 52-test automated unit, integration, and fuzz testing suite:

Test Category	Scope and Assertions	Result
Cryptographic Unit Tests	Canonical sorting, SHA-256 consistency, key-order invariance	12 / 12 Passed (100%)
Smart Contract Tests	CertificateRegistry deployment, issuance, zero-gas verify, revoke	28 / 28 Passed (100%)
Fuzz & Edge Tests	1-bit hash tampering, boundary timestamps, high-volume stress	12 / 12 Passed (100%)
TypeScript Static Check	Strict compiler type-checking (npx tsc --noEmit)	0 Errors (100% Valid)

8.2 Enhancements Beyond Minimum Requirements

- **1. Interactive Studio:** Live Split-Screen Diploma Preview Studio updating in real time as administrators type.
- **2. Multi-Modal Scanner:** Multi-Modal Verification supporting camera video feed, drag-and-drop QR images, and PDF document drops.
- **3. Audit Export:** Client-side CSV spreadsheet export for institutional record auditing.
- **4. Automated Email Delivery:** Automated SMTP email delivery with generated high-resolution vector PDF diploma attached.

9. Public GitHub Repository Link

The complete project codebase, smart contract source code, database migrations, automated tests, and documentation are publicly available on GitHub:

Repository Access: Public GitHub Repository URL:

<https://github.com/mengchheanglong/citadel-blockchain-certificates>

Branch: main | Status: Up to date with all 52 tests, smart contracts, and documentation.

10. Individual Contribution Report

This project was conducted as an Individual Capstone Project by Long Mengchheang. 100% of all technical planning, architecture, design, and implementation work was performed independently:

Technical Domain	Responsibilities & Deliverables	Contribution
Smart Contract & Web3	Wrote CertificateRegistry.sol, Hardhat deployment scripts, Ethers.js integration	100%
Frontend Engineering	Next.js 14 landing portal, dashboard, live preview studio, multi-modal scanner	100%
Backend & Database	Prisma PostgreSQL schema, Next.js API routes, Zod validation, PDF scan endpoint	100%
Document & Email Services	Vector diploma generation with embedded QR codes, automated SMTP dispatch	100%
Quality Assurance	52 automated tests, TypeScript strict compliance, diagrams, and project report	100%

11. Public Demo Video Link

A comprehensive demonstration video walking through the full lifecycle of the platform is accessible publicly at:

Video Access: Public Demonstration Video URL:

[Insert Public Google Drive or YouTube Link Here]

Access Permission: Public / Anyone with the link can view

Duration: Approximately 5 to 7 minutes

11.1 Demonstration Video Walkthrough Outline

Time	Stage / Feature	Demonstration Outline
0:00 - 0:45	Introduction	Overview of diploma fraud problem and Citadel's blockchain solution.
0:45 - 1:45	Dashboard	Walkthrough of live metric cards, credential table, and off-chain storage model.
1:45 - 3:00	Issuing Studio	Data entry, real-time diploma canvas rendering, and on-chain hash commitment.
3:00 - 3:45	Delivery	Inspection of generated vector PDF diploma with embedded QR code and email delivery.
3:45 - 5:15	Verification	Public zero-gas verification via camera scan, image drop, and PDF file upload.
5:15 - 6:30	Revocation & Summary	On-chain revocation demonstration with audit reason and concluding remarks.